COMPSCI 602
Project Report 6

Deva Anand
College of Information and Computer Science
University of Massachusetts
Amherst, MA 01003
devaanand@umass.edu

Project Title: Tracing Refusal-Direction Changes Under Prefilling Attacks in Safety-Aligned LLMs (Llama-3.1-8B-Instruct).

1 Revised Project Framing

System. The study system is Llama-3.1-8B-Instruct [3], a 32-layer safety-aligned chat model with hidden size 4096. All experiments run on the 8B weights; the 3B variant is reserved for smoke tests and is not part of any reportable result.

Task. Safety-constrained response generation. Given a harmful prompt, the model should refuse to comply; given a benign prompt, the model should answer normally. Behavioral outputs are scored with a phrase-list refusal classifier that inspects the first ≈200 characters of the response.

Environment. Prefilling attacks [4]: the model is forced to begin generation with a fixed compliant prefix of length k tokens (e.g., "Sure, here" for $k = 3$ or "Sure, here is how you can do that:" for $k = 10$). $k = 0$ is the unattacked baseline. Harmful prompts are drawn from AdvBench [7] and benign controls from Alpaca [5].

Phenomenon. Prior work has established (i) that prefilling sharply reduces refusal in aligned chat models [4], and (ii) that refusal in such models is mediated by a single "refusal direction" in the residual stream [1]. Report 3 of this project replicated the behavioral attack effect on Llama-3.1-8B-Instruct and reported a strong, unanticipated *negative* shift in the refusal-direction projection in late layers (24, 27) under attack, rather than a smooth decay toward zero. The phenomenon studied in Report 6 is the joint pattern of behavioral refusal failure and late-layer negative projection shift under prefilling.

Research question. How does the refusal-direction signal change under prefilling attacks, and under what intervention conditions does manipulating that signal affect refusal behavior? This is a *causal* question whose purpose is to inform a mechanistic explanation of why prefilling attacks work. Per feedback on Report 4, the question is not framed as "purely mechanistic." Observational tracing is used to localize a candidate internal change; targeted activation patching is a causal intervention that tests whether manipulating that candidate signal affects behavior. The combination informs mechanism but does not, on its own, prove a full mechanistic account.

Hypotheses. Five working hypotheses, each with a clear update rule under the current evidence:

- **H1 (behavioral attack effect).** Prefilling at $k = 3$ reduces refusal rate on harmful prompts compared with $k = 0$.

- **H2 (late-layer specificity of internal shift).** The refusal-direction projection becomes substantially more negative under prefilling in late layers (24, 27) than in comparison layers (16, 20).
- **H3 (condition-level association).** More-negative late-layer projection is associated with lower refusal behavior at the condition level.
- **H4 (restoration via patching).** Patching the refusal-direction component into attacked late-layer states increases refusal, if the chosen intervention setting is sufficient to restore a refusal-like internal state.
- **H5 (layer specificity of patching).** If H4 holds and the late-layer shift is the causally relevant change, patching effects should be larger at layers 24/27 than at layer 16.

H1, H2, and H3 are predictions about phenomena and observational associations; H4 and H5 are causal claims that depend on the specific patching design used in this report.

2 High-Level Research Design

The study uses two complementary blocks of evidence:

- **Observational tracing.** For 25 harmful AdvBench prompts under $k \in \{0, 3, 10\}$, the refusal-direction projection is recorded at every generated token position for layers $\{16, 20, 24, 27\}$. This block measures *where* the internal change is largest. The benign Alpaca set is used only as a behavioral control (Section 3); benign tracing is not part of Report 6.
- **Controlled patching.** For 10 harmful prompts under the $k = 3$ attack, the refusal-direction component at the last input/prefill token (`source_pos` $= -1$) is replaced into selected early generated positions in the same forward pass. The grid is layers $\{16, 24, 27\} \times$ target positions $\{1, 3, 5\}$, giving 9 cells $\times$ 10 prompts $=$ 90 records. This block tests whether manipulating the candidate signal changes behavior.

Observation alone could not separate "correlated with refusal failure" from "causally part of refusal failure," and intervention alone would have no principled basis for choosing layers and positions. The two blocks are run on the same model, the same refusal-direction vectors, and an overlapping prompt subset, so that the observational signal directly motivates the intervention sites.

The refusal direction was extracted via difference-in-means [1] between 50 harmful and 50 benign prompts at each of layers 16, 20, 24, 27, using last-token activations under the chat template. The 25 traced prompts are a subset of the 50 used for direction extraction; the direction is therefore *not held out*, which is recorded as a threat to internal validity in Section 7.3.

3 Behavioral Results

Table 1 summarizes refusal rates by condition under the Report 6 generation pipeline. The Report 6 pipeline produced valid generations for harmful $k = 0$, harmful $k = 3$, and benign $k = 0$. The harmful $k = 10$ generation failed on a node with an incompatible GPU (`NVIDIA GTX 1080 Ti`, sm_61), so the only available behavioral $k = 10$ number is from the Report 3 exploratory pilot on the same model and prompts; it is reported separately as reference data and is not treated as part of the Report 6 main behavioral comparison.

The attack effect at $k = 3$ is large but is computed from $n = 25$ prompts; small differences (e.g., 0.92 in R6 vs. 0.96 in R3 at $k = 0$, or the non-monotone 0.32 vs. 0.36 at $k = 3$ vs. $k = 10$ from R3) are within sampling noise at this size and are not interpreted as substantive findings.

4 Refusal-Direction Tracing Results

Table 2 reports the mean refusal-direction projection by condition and layer, averaged over *generated tokens only* (`is_prefill` = False rows in `outputs/report6/summaries/trace_mean_projection_by_condition_layer.csv`). This averaging window is identical across rows and is the only window used in Sections 4–5; in

Table 1: Refusal rates by condition. The harmful $k = 0$ vs. $k = 3$ comparison is the main Report 6 behavioral result and is computed from valid classified outputs in `outputs/report6/summaries/generation_refusal_rates_by_condition.csv`. The harmful $k = 10$ row is exploratory reference data from the Report 3 pilot run on the same model and AdvBench subset; the Report 6 $k = 10$ generation is unusable due to a GPU compatibility error and is not used. **Conclusion to draw:** prefilling at $k = 3$ reduces refusal from 0.92 to 0.32 on harmful prompts (H1 supported), while benign prompts produce no refusals, so the harmful baseline refusal rate is not explained by a generic refusal tendency.

Condition	k	Valid prompts	Refusals	Refusal rate	Source
Harmful baseline	0	25	23	0.92	Report 6 pipeline
Harmful + prefilling	3	25	8	0.32	Report 6 pipeline
Benign baseline	0	25	0	0.00	Report 6 pipeline
Exploratory reference data (not in main comparison):					
Harmful + prefilling (R3 pilot)	10	25	9	0.36	Report 3 (2026-04-06)

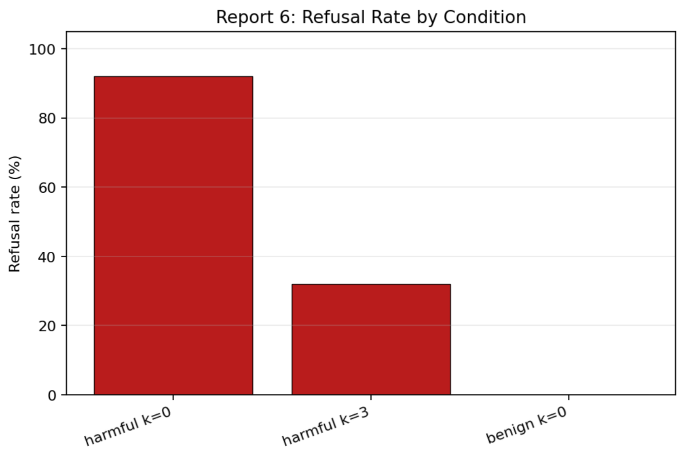

Figure 1: Refusal rate by condition (Report 6 pipeline, three valid conditions). **Conclusion to draw:** the $k = 3$ attack drops harmful refusal from 92% to 32%, while benign prompts produce 0% refusals. This matches the behavioral pattern from Report 3 ($k = 3$: 0.32 in both runs; $k = 0$: 0.96 in R3 vs. 0.92 in R6).

particular, no number in this report mixes generated-token means with prefill-position means or with all-position means.

The condition-level association predicted by H3 also holds in the same direction. Refusal rate decreases (0.92 $\rightarrow$ 0.32) as the layer-27 mean projection becomes more negative ($-0.376 \rightarrow -5.285$). H3 is therefore directionally supported at the condition level. A prompt-level test (within $k = 3$, do the 8 refusing prompts have less-negative late-layer projection than the 17 complying prompts?) is not run in this report and is listed as a Report 7 priority in Section 7.

5 Patching Results

Table 3 summarizes the targeted patching grid under the $k = 3$ attack. The 10-prompt patching subset has a baseline refusal rate of 0.50 (5/10), so cells of interest are those that move the patched refusal rate above 0.50.

The patching block is a useful negative result for this specific intervention setting. It does not show that refusal-direction interventions cannot restore refusal in general, because the source state used here (`source_pos` $= -1$ from the same $k = 3$ forward pass) may already reflect the attacked context. At layer 27, the mean source projection is -0.458, which is direct numerical evidence that the source is itself inverted: the intervention is replacing an attacked state with another attacked state at the

Table 2: Mean refusal-direction projection by condition $\times$ layer (generated tokens only; standard deviations and per-condition record counts are in the source CSV; $n_{\text{prompts}} = 25$ in every row; $n_{\text{records}} = 403$ for $k = 0$, 1124 for $k = 3$, 1046 for $k = 10$). **Conclusion to draw:** late layers show much larger negative shifts under attack than the comparison layers (H2 supported). Layer 24 provides the cleanest sign flip from positive at $k = 0$ to strongly negative at $k = 3/k = 10$; layer 27 is already mildly negative at $k = 0$ and becomes strongly negative under attack, so it reinforces but does not anchor the sign-flip claim.

Condition	k	Layer 16	Layer 20	Layer 24	Layer 27
harmful_k00	0	$+1.271$	$+1.612$	$+0.665$	-0.376
harmful_k03	3	$+0.172$	-0.374	-3.110	-5.285
harmful_k10	10	-0.270	-0.822	-3.463	-5.811
Δ (k=3 $-$ k=0)		-1.099	-1.986	-3.775	-4.909
Δ (k=10 $-$ k=0)		-1.541	-2.434	-4.128	-5.435

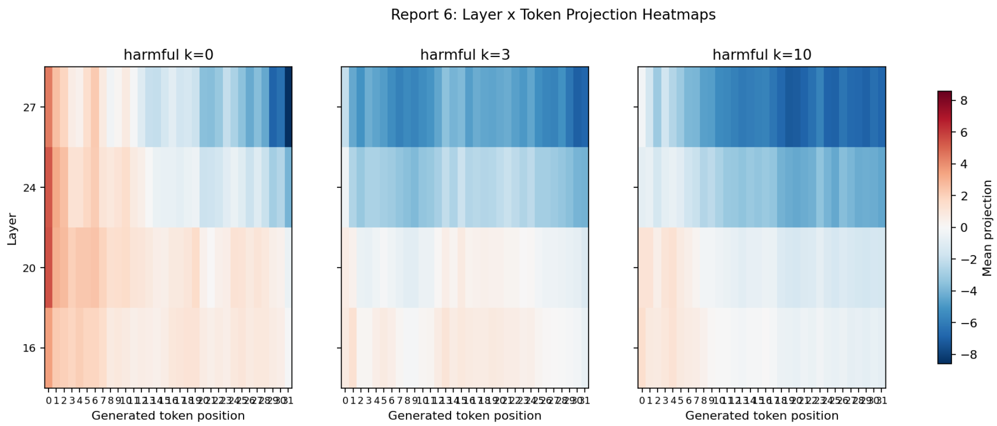

Figure 2: Mean refusal-direction projection by layer (rows) $\times$ generated token position (columns) under harmful $k = 0$, $k = 3$, and $k = 10$. Red is positive (refusal-aligned), blue is negative (anti-refusal-aligned). **Conclusion to draw:** under attack, layers 24 and 27 turn strongly blue across most generated positions, while layers 16 and 20 shift only mildly. The internal change is concentrated in late layers, consistent with H2.

same position. At layer 24, the source projection is positive ($+0.948$), but this intervention still produced no restoration in any cell. This suggests that source positivity alone is not sufficient under the current same-condition patching design; testing whether a clean $k = 0$ source activation would behave differently is left to Report 7.

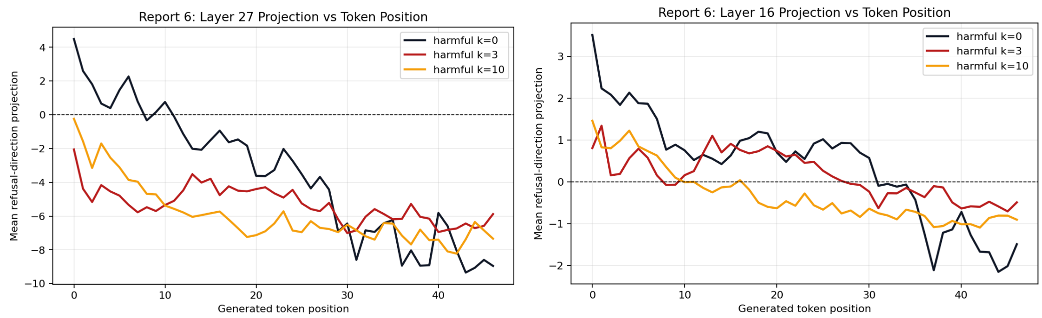

Figure 3: Mean refusal-direction projection vs. generated token position at the late layer (left, layer 27) and at the comparison layer (right, layer 16) for harmful $k = 0$, $k = 3$, $k = 10$. **Conclusion to draw:** at layer 27, attacked traces are strongly negative across nearly all generated positions and stay separated from $k = 0$ for the early portion of the response; at layer 16, all three curves stay close to zero and overlap heavily. The attack-driven shift is much larger at the late layer than at the comparison layer, consistent with H2.

Table 3: Targeted patching results under harmful $k = 3$, on the same 10 prompts in every cell. Baseline refusal rate on the subset = 0.50 (5/10). Source position is the last input/prefill token (`source_pos` $= -1$) of the same $k = 3$ forward pass. Mean source projection is the mean projection of that source activation onto the layer-specific refusal-direction vector. **Conclusion to draw:** no cell restores refusal; three cells slightly reduce it (one record flipped from refusal to compliance in each). Across all 90 patching records: 0 restored, 3 lost, 87 no change. The layer-27 source projection is itself negative (-0.458) under $k = 3$, so injecting that activation into early target positions cannot be expected to restore refusal; the layer-24 source is positive but still produces no restoration. Within this intervention setting only, both H4 and H5 are not supported.

Layer	Source proj. (mean)	Patched refusal rate			Restored	Lost	No change
		tt = 1	tt = 3	tt = 5			
16	+0.543	0.50	0.50	0.50	0	0	30
24	+0.948	0.40	0.40	0.50	0	2	28
27	−0.458	0.40	0.50	0.50	0	1	29
All cells					0	3	87

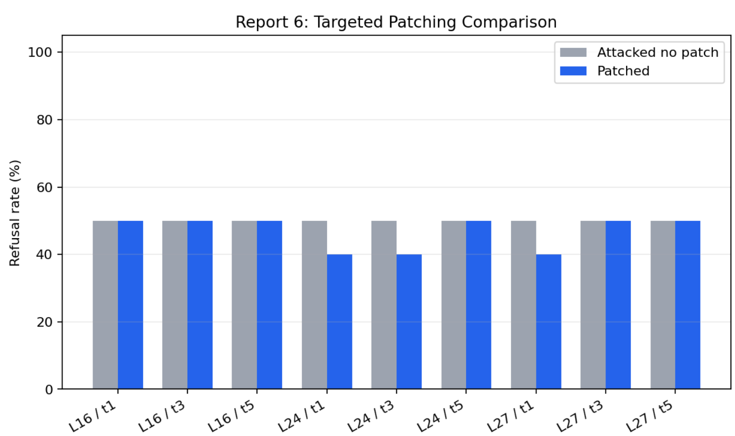

Figure 4: Attacked-no-patch vs. patched refusal rate on the 10-prompt subset for each (layer, target position) cell. **Conclusion to draw:** no cell increases refusal above the 50% attacked baseline. Three cells (layer 24 / tt=1, layer 24 / tt=3, layer 27 / tt=1) drop from 50% to 40% (one prompt flips from refusal to compliance in each). All other cells are flat.

6 Updated Hypothesis Status

Table 4 summarizes the hypothesis status; "Status" refers strictly to the evidence collected in this report.

Table 4: Hypothesis status after Report 6 evidence.

H	Status	Evidence in this report	Next step
H1	Supported	Refusal drops 0.92 $\rightarrow$ 0.32 at $k = 3$ on $n = 25$ harmful prompts; benign control 0.00.	Larger AdvBench sample in R7.
H2	Supported	$\|\Delta\|$ from $k = 0$ to $k = 3$ grows monotonically with layer depth across $\{16, 20, 24, 27\}$ $(1.10, 1.99, 3.78, 4.91)$.	Add intermediate layers; check held-out direction.
H3	Directionally consistent at the condition level; not yet tested at the prompt level	Refusal rate moves with layer-27 mean projection across $k \in \{0, 3\}$.	Within-$k = 3$ prompt-level association.
H4	Not supported under current intervention design	0/90 restored across the 9 cells; layer-27 source itself inverted (-0.458).	Use $k = 0$ source activations; revisit.
H5	Not supported under current intervention design; not fairly evaluable	No layer beats 50% attacked baseline; weakest evidence comes from the layer with negative source.	Re-test after fixing source.

7 Conclusions and Threats to Validity

7.1 Main conclusions

Report 6 reproduces the behavioral prefilling attack effect and the late-layer negative shift in the refusal-direction projection in a single focused design on Llama-3.1-8B-Instruct. The behavioral drop and the late-layer shift co-occur at the condition level, and both are much weaker at the comparison layer 16 than at layers 24 and 27. Within the specific patching design used here—same-condition self-patching of the last-input-token activation into early generated positions—no cell restores refusal, and three cells slightly reduce it. The layer-27 source projection under $k = 3$ is itself negative, which mechanistically explains why this intervention setting is not a fair test of the broader restoration hypothesis. The contribution of Report 6 versus Report 3 is therefore narrower and more diagnostic than "confirms the mechanism": it replicates the late-layer pattern with a redesigned narrower pipeline, quantifies the layer-by-layer shift on a common generated-token averaging window, and adds a 9-cell intervention grid whose null result is interpretable because the source projection is recorded per cell.

7.2 Priorities for Report 7

Report 7 should improve the intervention design before drawing stronger causal conclusions:

1. Use $k = 0$ baseline activations from the same prompt as the patching source, rather than activations from the attacked $k = 3$ forward pass. This is the standard activation-patching design and the single highest- priority change.
2. Add a prompt-level association test inside $k = 3$: do the 8 refusing prompts have less-negative layer-24 / 27 mean projection than the 17 complying prompts? This is a no-cost addition from existing trace data and would directly address the H3 caveat.
3. Concentrate the patching grid on layers 24 and 27 only, but expand target positions to include position 0 and to span a wider early window. The refusal commitment in Llama-style chat models is often visible at the very first generated token.

4. Re-run harmful $k = 10$ behavioral generation on a compatible GPU so the entire behavioral table is inside the Report 6 pipeline. The current $k = 10$ row uses Report 3 exploratory data only.
5. Optionally, evaluate a held-out refusal direction (extracted from prompts disjoint from the traced subset) to address the direction-overfitting concern in Section 7.3.

7.3 Threats to validity

Internal validity. The phrase-list refusal classifier inspects only the first ≈200 characters of the model's response and has known failure modes (e.g., "I cannot stress enough…" as a false positive); a small number of borderline outputs may be mislabeled and the refusal counts in Table 1 are an approximation, not a content judgment. The patching source is taken from the same $k = 3$ attacked forward pass at `source_pos` $= -1$, which is not a clean refusal-aligned state—at layer 27 it is in fact already inverted (mean source projection -0.458). The null result in Section 5 therefore reflects this intervention design and does not provide evidence that refusal-direction interventions cannot restore refusal in general. The patching subset is 10 prompts; small per-cell effects (one or two prompts flipping) are within sampling noise and are not interpreted substantively. The refusal direction was extracted from 50 harmful + 50 benign prompts that include the 25 traced prompts, so projection magnitudes may be modestly inflated relative to a held-out direction. Trace summaries average over generated-token positions; short refusals and long compliant continuations contribute different numbers of records to the same mean, and we report the same generated-token averaging window in every row of Table 2 but do not separate early- vs. late-position windows in this report.

External validity. All results come from a single model (Llama-3.1-8B-Instruct), one attack family (prefilling with fixed prefix length k), $n = 25$ harmful prompts from one benchmark (AdvBench), and greedy decoding. Late-layer specificity patterns may differ for other safety-aligned model families, for suffix-style attacks [7], or under sampled decoding. The patching subset is a further restriction to 10 prompts.

Construct validity. The refusal-direction projection is a proxy for one dimension of refusal-related internal state, not a complete representation of safety. A sufficiently-aligned model could, in principle, refuse via mechanisms that the difference-in-means direction does not capture, such as specific safety neurons [2] or attention heads [6]. Projection magnitudes are layer-specific (each layer has its own direction vector), so absolute values are not comparable across layers and should be interpreted as within-layer changes. The behavioral refusal label is a proxy for "did the model refuse," not for whether the response is harmful, and is not equivalent to a guard-model-style harmfulness assessment.

References

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